



OPEN Walnut-shell-derived porous carbon for efficient room-temperature adsorption of organic dyes

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Sustainable porous carbons derived from agricultural waste represent promising materials for wastewater remediation. In this study, activated carbons were synthesized from walnut shells via a single-step carbonization–chemical activation process using KOH and K_2CO_3 at 700 and 800 °C. The optimal material, WS- K_2CO_3 -800 °C, exhibited a high specific surface area of 1786.64 m² g⁻¹ and a hierarchical pore structure, as confirmed by N₂ adsorption–desorption analysis. Its adsorption performance was evaluated for methylene blue (MB), methyl orange (MO), and Congo red (CR) at room temperature. Rapid adsorption was observed, achieving near-complete dye removal within 30 min at initial concentrations of 20 ppm for MB and MO, and 10 ppm for CR. Kinetic modelling showed excellent agreement with the pseudo-second-order model, with correlation coefficients (R^2) of 0.998, 0.996, and 0.995 for MB, MO, and CR, respectively, indicating chemisorption-dominated uptake. The calculated equilibrium adsorption capacities reached 185.8 mg g⁻¹ for MB, 133.7 mg g⁻¹ for MO, and 88.9 mg g⁻¹ for CR. Isotherm analyses indicated predominantly heterogeneous and multilayer adsorption behaviour, with the Freundlich and Temkin models providing the best fits. The superior performance of WS- K_2CO_3 -800 °C is attributed to its high surface area, hierarchical porosity, and surface functionality, highlighting walnut-shell-derived activated carbon as a low-cost, renewable, and effective adsorbent for dye-contaminated wastewater.

Keywords Walnut shell, Activated carbon, Adsorption, Organic dye removal, Wastewater treatment, Sustainable materials

The water treatment industries are the main drivers of the demand for activated carbon^{1,2}. Rapid population expansion, improved living standards, and a rising awareness of environmental pollution have intensified the need for cleaner water and air, consequently increasing the consumption of activated carbon. At the same time, the increasing production and use of dyes, particularly in the textile sector, further elevate this demand^{3–6}.

Dyes, whether synthetic or natural, are chemical compounds designed to bind to surfaces and impart colour⁷. Most commercial dyes are structurally complex and engineered to resist detergents, ultraviolet radiation, temperature variations, and bleaching agents. Their applications span multiple industries, including textiles, paper, leather tanning, food processing, plastics, cosmetics, printing, and wastewater treatment. Globally, more than 100,000 commercial dyes are produced, amounting to nearly 700,000 tonnes per year^{8–10}. The textile sector alone consumes over 10,000 tonnes annually, with an estimated 100 tonnes ultimately discharged into aquatic environments. The widespread release of synthetic dyes has therefore become a significant environmental concern, particularly because many dyes are toxic, persistent, and resistant to conventional treatment methods. Azo dyes (such as methylene blue, methyl orange, and Congo red (Fig. 1)) are among the most hazardous dye categories due to the toxic aromatic amines released during their degradation^{11–16}. These dyes are linked to mutagenicity, chromosomal abnormalities, carcinogenicity, and respiratory toxicity^{17–21}. Similarly, anthraquinone and reactive

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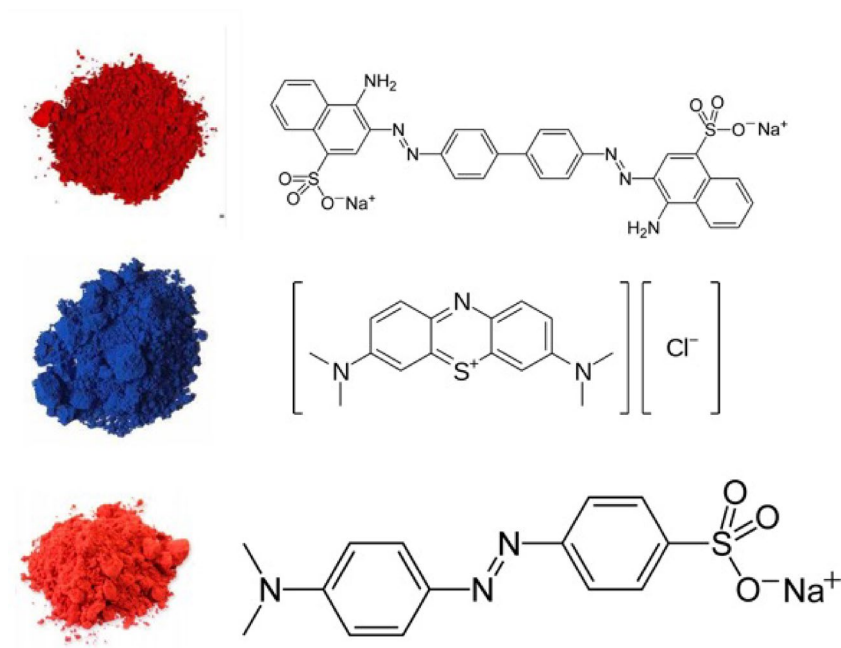


Fig. 1. Typical chemical configuration of the investigated Azo-based dyes; Methylene blue, Methyl orange and Red Congo⁸.

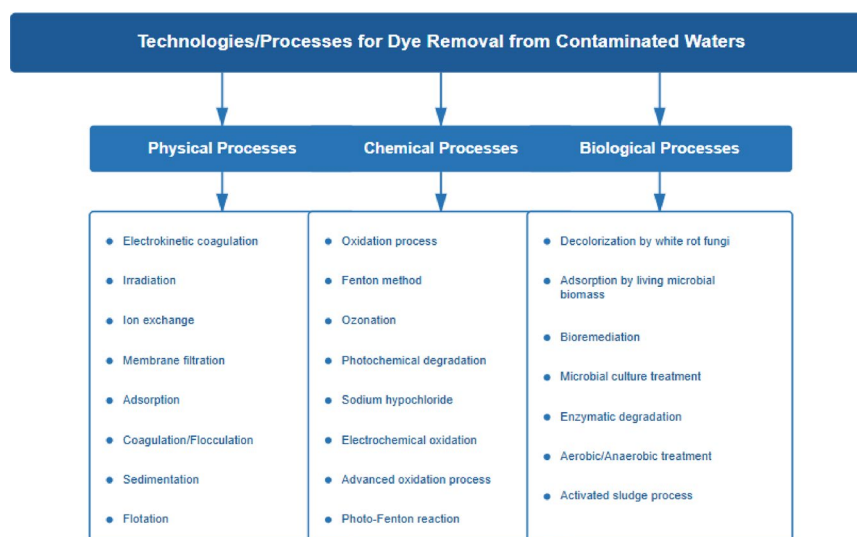


Fig. 2. Major processes for dyes removal⁷.

dyes display strong chemical stability and low biodegradability, enabling them to persist in effluents for long periods, thus demanding more effective removal strategies^{22–24}.

A variety of physical, chemical, and biological methods are used for dye removal, as illustrated in Fig. 2. Among these techniques, adsorption stands out as one of the most efficient and economically attractive processes for removing organic and inorganic contaminants from industrial effluents⁹. This method relies on solid adsorbents to capture dye molecules through either physisorption (weak physical forces) or chemisorption (strong chemical bonding)^{10,25–27}. Due to its simplicity and high removal efficiency, adsorption, particularly using activated carbon, remains the preferred treatment method in the textile and related industries²⁸.

The high cost of commercial activated carbon has motivated extensive research into low-cost, sustainable, and high-performance alternatives. Numerous studies have explored adsorbents derived from natural materials, agricultural residues, industrial by-products, and biosorbents (Table 1)^{29–34}. Agricultural biomass, rich in lignocellulosic components, is especially attractive because it provides renewable carbonaceous precursors with good adsorption potential^{35,36}. Various physical and chemical treatments have been used to enhance the porosity,

References	Activation/treatment method	Dye type	Results (reported q_m /removal)
Uddin et al. ²⁴	Raw walnut shell powder (characterization and optimization; no high-temp chemical activation)	Methylene Blue	Good MB removal in batch tests; detailed isotherm/kinetic analysis (study shows WS is a promising low-cost adsorbent)
Farch et al. ⁴¹	Raw walnut shell powder (batch adsorption study; process optimization)	Methylene Blue	Reported optimized MB removal under varying pH/concentration; adsorption capacity and process parameters presented
Zhuang et al. ⁴²	KOH activation (KOH-activated walnut shell biochar)	Methylene Blue	KOH activation markedly increased adsorption capacity and improved kinetics; full characterization (BET, SEM, FTIR) (q_m)
Tunay et al. ⁴³	H ₃ PO ₄ activation (phosphoric acid-activated WSAC)	(includes dye adsorption tests, e.g., MB/others; may include CR)	Phosphoric-acid activation produced high-performance WSAC for organic contaminants; detailed adsorption metrics provided (high removal percentages)
Rose et al. ⁴⁴	Amino-functionalization (cationic modification)	Congo Red	$Q_e \approx 44.36 \text{ mg g}^{-1}$, 94.12% removal reported for cationic amino-modified walnut shell (CMWS)—shows surface charge modification greatly improves CR uptake
Ojo et al. ⁴⁵	H ₃ PO ₄ functionalization/acid activation	Congo Red	Reported $\sim 94.5\%$ CR removal at acidic pH; Langmuir q_m reported $\approx 40 \text{ mg g}^{-1}$ in their conditions
Kahoul et al. ⁴⁶	Microwave-induced KOH activation	Methylene Blue	KOH activation (microwave) produced high surface area activated carbons and significant dye removal; useful process-engineering reference
McCaffrey et al. ⁴⁷	Chemical activation/lab-scale AC synthesis (comparison to commercial ACs)	Methylene Blue	Walnut-derived AC showed adsorption comparable to commercial activated carbons (practical benchmark)
Kuyucu et al. ⁴⁸	KOH activation (1:3) yielding very high BET (reported $\sim 2347 \text{ m}^2 \text{ g}^{-1}$)	Reactive dyes (Reactive Blue 19, Reactive Red 195)	Shows KOH activation yields exceptional surface area and strong dye adsorption (example for reactive dyes; method transferable to CR/MB)
Liu et al. ⁴⁹	Raw walnut shell	Methylene Blue (MB) and Congo Red	Reported efficient MB removal and MB-loaded WS to remove Congo red

Table 1. Major published literature on dyes removal by activated carbon derived from Walnut shells.

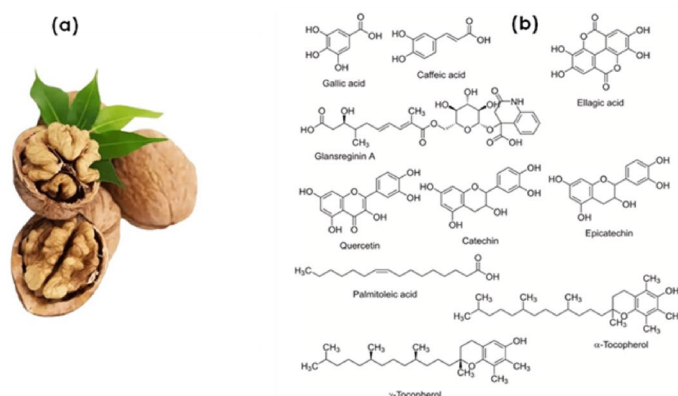


Fig. 3. (a) Typical configuration of standard Walnut fruit with its kernel and shell, (b) Major phyto-compound of the natural extract of the Walnut kernel⁴².

surface chemistry, and adsorption efficiency of these biomass-derived adsorbents^{11–13,37–40}. While earlier biomass-derived adsorbents demonstrated promise, they often required multi-step chemical modifications (e.g., esterification with citric acid, nanoparticle incorporation and polymer grafting, or sequential amine functionalization) that increase processing complexity, chemical consumption, and cost.

In this context, the present study introduces, for the first time, a nano-scaled graphitic adsorbent synthesized from walnut shell biomass (Fig. 3a). Walnut shells are a promising precursor not only due to their lignocellulosic composition but also because walnut kernels contain rich phytochemical profiles (Fig. 3b) known to support the green synthesis of nanomaterials^{41–48}. These bioactive compounds include gallic acid, caffeic acid, ellagic acid, quercetin, catechin, epicatechin, glansreginin A, α-tocopherol, and related molecules⁴⁹.

The engineered nano-graphitic material derived from walnut shell demonstrates multi-adsorptive capabilities toward several toxic amine-based dyes^{50–59}. As illustrated in Table 1, previous studies have employed raw walnut shell, chemically activated biochar, and functionalized composites for dye removal; however, none have reported the bioengineering of nano-scaled graphitic carbon from walnut shell with the exceptionally high adsorption efficiencies observed in this work. The walnut-shell-derived activated carbons produced in this study via single-step KOH/K₂CO₃ activation at 700–800 °C exhibited exceptionally high surface areas (up to 1786.64 m² g^{−1}) and hierarchical pore structures. This streamlined synthesis reduces the number of production steps, minimizes chemical inputs, and lowers energy and resource demands, thereby enhancing sustainability. The combination of simplicity and efficiency not only improves scalability but also aligns with circular economy principles by valorizing agricultural waste into high-performance adsorbents. Moreover, the materials achieved rapid dye removal within 30 min at low concentrations, outperforming earlier adsorbents that required longer

equilibrium times. Thus, the sustainability of the single-step approach lies in its ability to reduce processing complexity while delivering superior adsorption capacity and kinetics, positioning walnut-shell activated carbon as a cost-effective and environmentally responsible solution for wastewater remediation.

Experiments and methods

Reagents and materials

The Walnut shells were obtained from a market in Tehran (Iran). All chemicals used in this study were of analytical grade and used without further purification. Potassium carbonate (K_2CO_3 , $\geq 99\%$) and Potassium hydroxide (KOH, $\geq 99\%$) were purchased from LabChem. Hydrochloric acid (37% w/w, ACS reagent grade) was also purchased from LabChem. Deionised water was obtained using a Deionized water purification station. All reagents met the analytical grade specifications required for the experimental procedures described in this work.

Synthesis of WS AC

Walnut shells were thoroughly washed with deionized water, treated with ethanol to remove surface impurities, and dried at 60 °C for 12 h. The dried shells were then pulverized using a vibrating ball mill (260 W) for 15 min to obtain fine homogeneous particles suitable for activation. For each batch, 10 g of the crushed walnut shells were mixed with 10 g of activating agent, either potassium hydroxide (KOH) or potassium carbonate (K_2CO_3), maintaining a 1:1 mass ratio of precursor to activating agent. The mixture was homogenized with 5 mL of deionized water to form a uniform paste, then dried at 60 °C for 12 h to ensure complete impregnation. The dried composites were carbonized and activated simultaneously in a tube furnace under a continuous argon flow ($200\text{ cm}^3\text{ min}^{-1}$), with a heating rate of 5 °C min^{-1} . Activation was conducted at 700 °C or 800 °C for 2 h, yielding four distinct samples designated as: WS-KOH 700 °C, WS-KOH 800 °C, WS- K_2CO_3 700 °C, WS- K_2CO_3 800 °C. After thermal treatment, the resulting carbons were washed with 3 M HCl for 5 h to remove residual inorganic species, followed by repeated rinsing with deionized water until the filtrate reached neutral pH. The purified samples were finally dried at 60 °C for 12 h and stored in airtight containers for further characterization. The synthesis parameters were selected based on prior optimization from related biomass-derived systems (e.g., MS AC).

Characterization

N_2 adsorption-desorption isotherms were determined at 77 K using a NOVA-touch LX²-Quanta Chrome instrument to evaluate textural features via the Brunauer-Emmett-Teller method (relative pressure: 0.05–0.95). The morphology was examined through scanning electron microscopy on a Zeiss Ultra Plus 55 FE-SEM operating at 2.0 kV. Vibrational modes were analysed using a WITec Alpha 300 RAS + Raman spectrometer ($\lambda = 523\text{ nm}$, 4 mW laser power, 120 s integration time).

Adsorption calculations

Dye adsorption experiments were conducted to evaluate the purification performance of WS- K_2CO_3 800 °C material using three model dyes: methyl orange, methylene blue, and Congo red. Dye concentrations were determined spectrophotometrically using the Beer-Lambert Law; $A = \xi\delta\chi$, where A represents the measured absorbance, ξ is the molar absorptivity coefficient, δ is the optical path length, and χ is the molar concentration⁵⁰. For methyl orange, a molar absorptivity coefficient of $7.4 \times 10^4\text{ L mol}^{-1}\text{ cm}^{-1}$ at $\lambda_{\text{max}} = 464\text{ nm}$ and a molecular weight of 327.33 g mol^{-1} were employed. For methylene blue, a molar absorptivity coefficient of $9.5 \times 10^4\text{ L mol}^{-1}\text{ cm}^{-1}$ at $\lambda_{\text{max}} = 664\text{ nm}$ and a molecular weight of 319.85 g mol^{-1} were considered⁶⁰. For Congo red, a molar absorptivity coefficient of $4.5 \times 10^4\text{ L mol}^{-1}\text{ cm}^{-1}$ at $\lambda_{\text{max}} = 497\text{ nm}$ and a molecular weight of 696.66 g mol^{-1} were applied. All spectrophotometric measurements were conducted using standard $\delta = 1\text{ cm}$ path length quartz cuvettes to ensure consistent optical conditions²⁹. Applying Beer's Law to $\xi = A/\delta\chi$. The concentrations were then converted to mass concentrations (ppm) using the relationship of $C\text{ (ppm)} = \chi\text{ (mol/l)} \times \text{MW (g mol}^{-1}) \times 10^{-3}$. The linear relationship between absorbance and concentration was validated across the working range ($R^2 > 0.99$) for all three dyes, confirming adherence to Beer-Lambert Law assumptions. The relative adsorption efficiencies of WS- K_2CO_3 800 °C were evaluated using various concentrations of aqueous solutions of methyl orange, methylene blue, and Congo red under identical experimental conditions to establish the optimal adsorption performances. All experiments were conducted at ambient temperature ($25 \pm 2\text{ °C}$) with continuous agitation. UV-visible absorption measurements were recorded at predetermined intervals (0–60 min) to monitor adsorption kinetics. Residual dye concentrations were quantified using pre-established calibration curves for each dye, and adsorption capacities were calculated from the difference between initial and final concentrations:

$$q_t = \frac{C_o - C_t}{m} \times V \quad (1)$$

where q_t is the adsorption capacity at time t in mg g^{-1} , C_o is the initial dye concentration (ppm), C_t is the dye concentration at time t (ppm), V is the volume of methyl orange solution (L) and m is the mass of adsorbent (g).

The adsorption at equilibrium (q_e) was determined as follows⁵²:

$$q_e = \frac{C_o - C_e}{m} \times V \quad (2)$$

where q_e is the adsorption capacity at equilibrium (mg g^{-1}), C_o is initial dye concentration (ppm), C_e is the dyes equilibrium concentration (after 24 h of adsorption) (ppm), V is the volume of methyl orange solution (L) and m is the mass of adsorbent (g).

The dyes rate of removal efficiency (R = removal efficiency (%)) was determined as follows⁵²:

$$R(\%) = \frac{C_o - C_t}{C_o} \times 100 \quad (3)$$

Reproducibility

All experiments were conducted in triplicate ($n \geq 3$). Standard deviations were $< 3\%$ for BET surface area, and $\pm 2\%$ for adsorption measurements. Synthesis reproducibility across multiple batches showed performance parameters consistently within $\pm 5\%$, confirming protocol robustness and scalability.

Results and discussions

Synthesis procedure

Figure 4 illustrates the developed green and sustainable carbonization/activation method used to produce nanoscale porous carbons from walnut shells. This method consists of a pyrolysis combining carbonization and the activation process. As compared to previous productions of walnuts shells-activated carbon^{53–56}, the method did not require prior pre-carbonization to obtain hierarchical and highly porous activated carbon. This is, therefore, advantageous as it promotes sustainable technology by reducing the activated carbon production steps. The percentages of yield of carbon obtained using KOH, K_2CO_3 were 32%, 31%, 30%, and 36%, for the WS-KOH 700 °C, WS-KOH 800 °C, WS- K_2CO_3 700 °C, and WS- K_2CO_3 800 °C, respectively. It is suggested that the presence of K_2CO_3 within the core of the precursor restricted the formation of tar and additional liquids such as methanol and acetic acid by forming cross-links, and obstructing the shrinkage of the precursor particle by occupying substantial volumes^{57,61–65}.

This supports previous findings suggesting that K_2CO_3 only reacts with the char and gets removed during the gasification process. Consequently, the process occurs in an inert atmosphere as follows^{61,64}:



The activation process using KOH is likely to occur as follows⁵⁹:



Figure 5 presents the nitrogen adsorption and desorption isotherms of the synthesized samples produced under the four distinct synthesis conditions. According to the 2015 IUPAC classification, all samples exhibit type IV a isotherm accompanied by H4 hysteresis loops, which are indicative of materials that contain micro-pores combined with narrow slit shaped meso-pores⁶⁶. This combination of pore structures improves adsorption behavior by providing a large accessible surface area while facilitating efficient molecular transport⁶⁷. The specific surface areas obtained for WS KOH at 700 °C, WS KOH at 800 °C, WS K_2CO_3 at 700 °C, and WS K_2CO_3 at 800 °C are 1082.64, 1230.96, 1261.83, and 1786.64 square meters per gram, respectively. These values, together with the adsorption capacities and pore volume parameters summarized in Table 2, clearly illustrate how activation chemistry and temperature govern the development of adsorption-relevant textural properties. Notably, WS K_2CO_3 at 800 °C shows the highest surface area and pore volume, which directly correlates with its enhanced adsorption performance. This improvement is attributed to the combined effects of potassium carbonate activation and elevated thermal treatment, which promote extensive pore development and restructuring of the carbon framework^{61,64}. The pore size distribution presented in Fig. 5b confirms the presence of broader pore

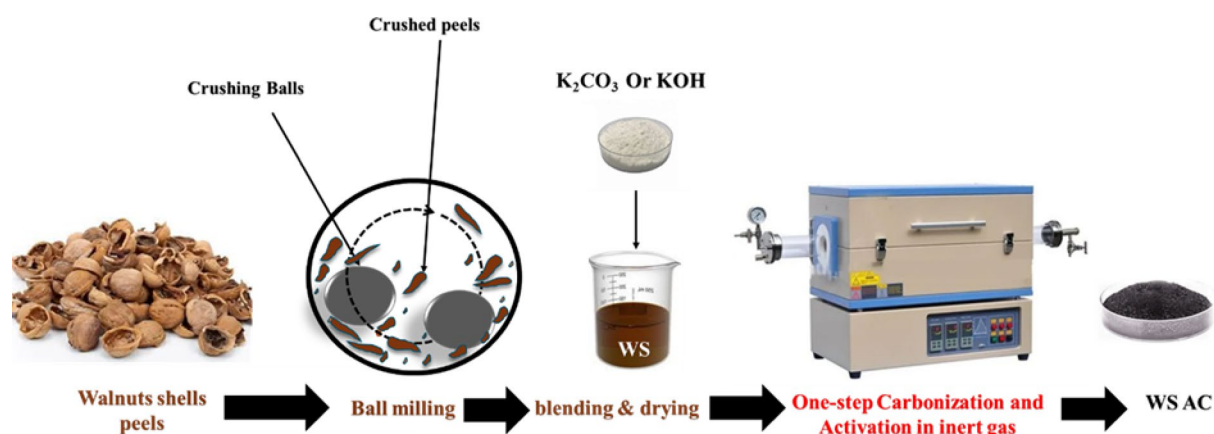


Fig. 4. Single-pot synthesis of walnut-shell-derived activated carbons. The figure outlines the one-step process where milled walnut shell biomass is impregnated with KOH or K_2CO_3 and thermally treated under inert conditions, enabling simultaneous carbonization and activation at 700 °C or 800 °C. Washing and drying yield porous carbons whose properties depend on the activating agent and temperature.

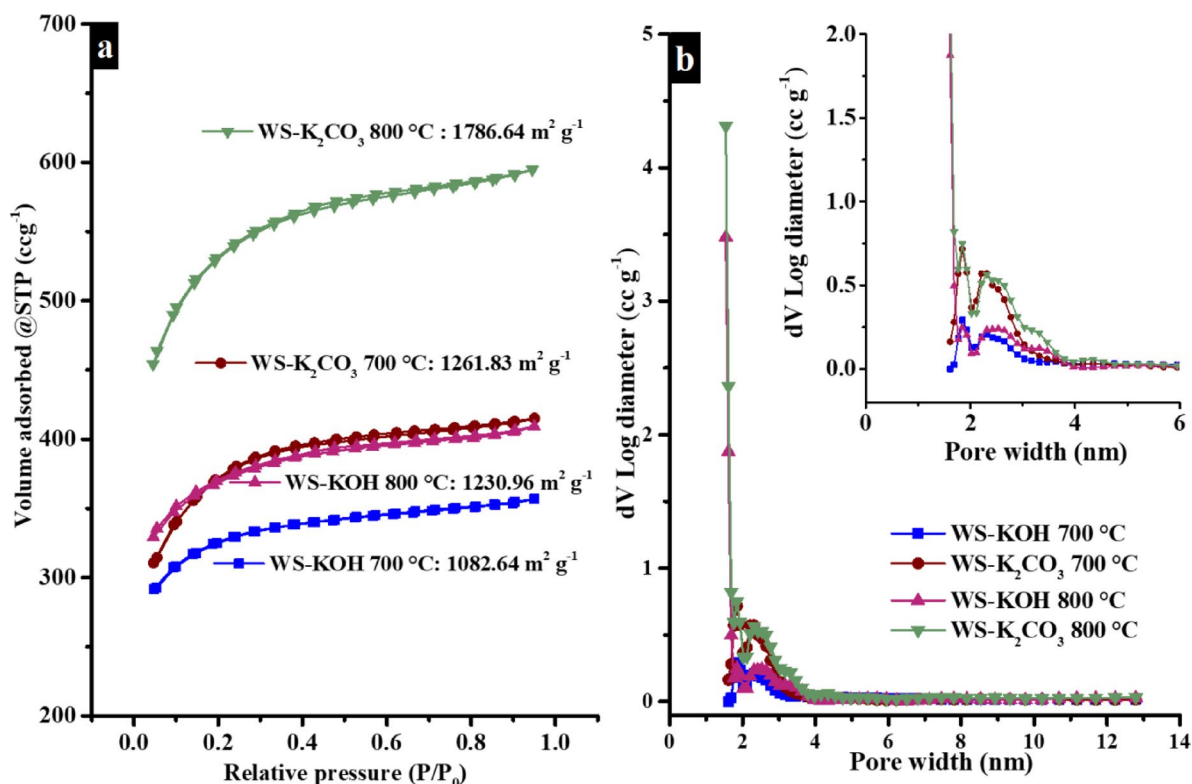


Fig. 5. Textural properties of the activated carbons. (a) N₂ adsorption–desorption isotherms showing type IVa behaviour with H4 hysteresis, indicating micropores and narrow mesopores. (b) Corresponding pore-size distributions highlighting micro–mesoporous structures, with WS-K₂CO₃ 800 °C displaying the widest pore development.

Materials	WS-KOH 700 °C	WS-KOH 800 °C	WS-K ₂ CO ₃ 700 °C	WS-K ₂ CO ₃ 800 °C
Micropore volume (cc g ⁻¹)	0.45	0.49	0.48	0.69
Total pore volume (cc g ⁻¹)	0.55	0.63	0.64	0.92
Average pore size (nm)	2.04	2.06	2.04	2.06

Table 2. Textural properties of all the synthesized graphite samples.

channels in WS K₂CO₃ at 800 °C, reflecting a hierarchical porous architecture. The coexistence of micro-pores and meso-pores allows rapid diffusion of adsorbate molecules and high uptake capacity, making this material particularly well-suited for adsorption-based applications such as contaminant removal and gas capture^{68,69}.

Morphological studies

Figure 6 presents the High-resolution scanning electron microscopy image at low-magnification (1 μm) and high-magnification (200 nm) of the optimized nano-scaled porous carbon produced under WS K₂CO₃ at 800 °C synthesis conditions. The micrograph displays a clear graphitic foam-like structure consisting of interconnected carbon walls and numerous pore channels. This porous and coherent architecture verifies the successful formation of graphitic carbon and emphasizes the high porosity of the bioengineered material.

Vibrational optical investigations

Figure 7 presents the room-temperature Raman spectrum of the optimized porous carbon (WS K₂CO₃ at 800 °C). The spectrum exhibits two relatively sharp Raman-active modes at approximately 1349 cm⁻¹ and 1583 cm⁻¹, corresponding to the characteristic D and G bands of graphitic carbon, respectively. The G band (~1580 cm⁻¹) arises from the in-plane vibrational mode of sp²-bonded carbon atoms in the graphite lattice, while the D band (~1350 cm⁻¹) is associated with structural defects, edge sites, and lattice distortions. Additionally, a broader feature appears near 2883 cm⁻¹, which can be assigned to the combined second-order (2D) overtone modes typical of graphitic materials. Taken together, these Raman features confirm the successful formation of graphitic carbon within the bio-engineered material derived from walnut shell.

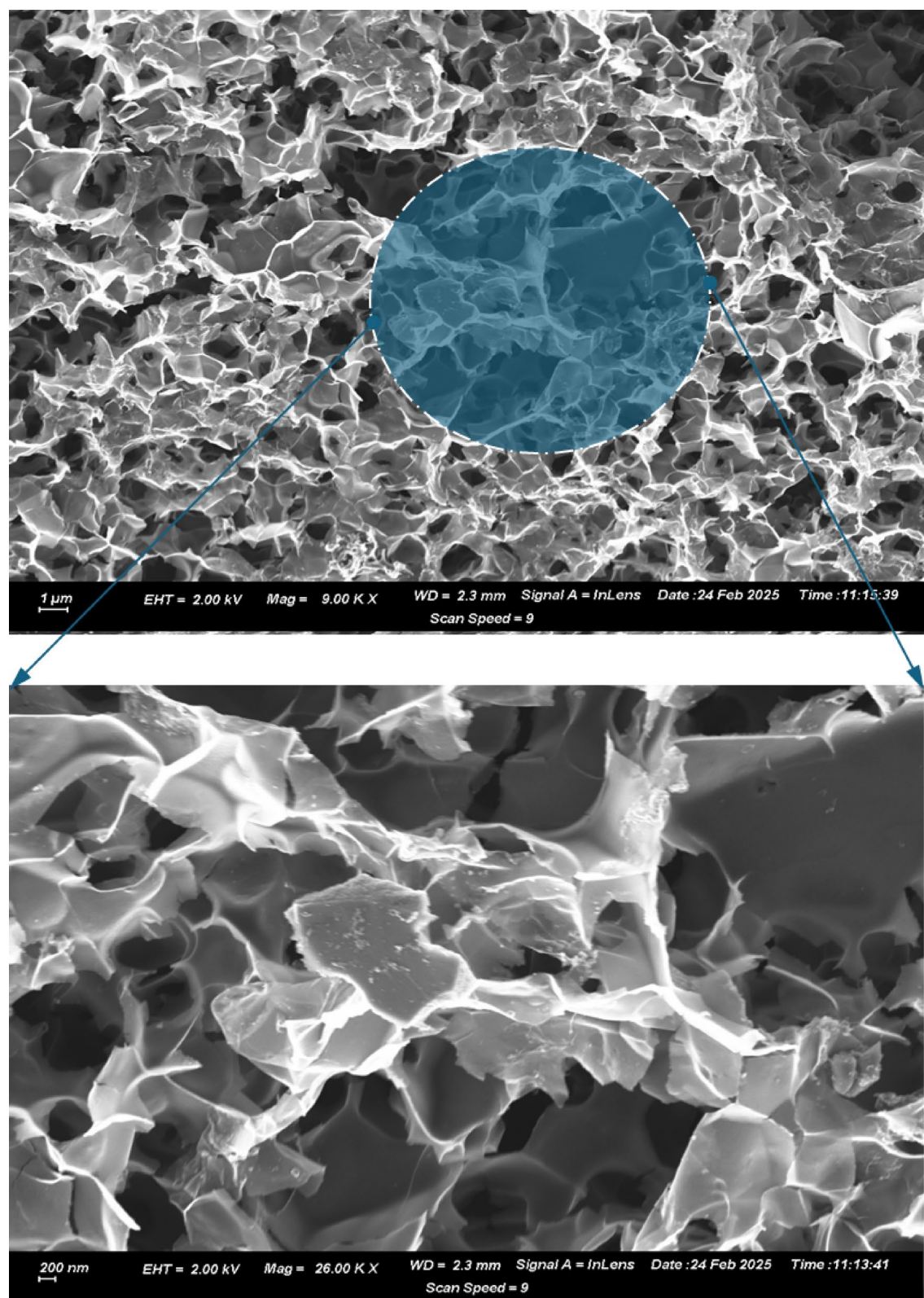


Fig. 6. Morphology of WS- K_2CO_3 800 °C porous carbon. High-resolution SEM images at 1 μm and 200 nm magnification reveal the interconnected, highly porous graphitic framework of the WS- K_2CO_3 800 °C material.

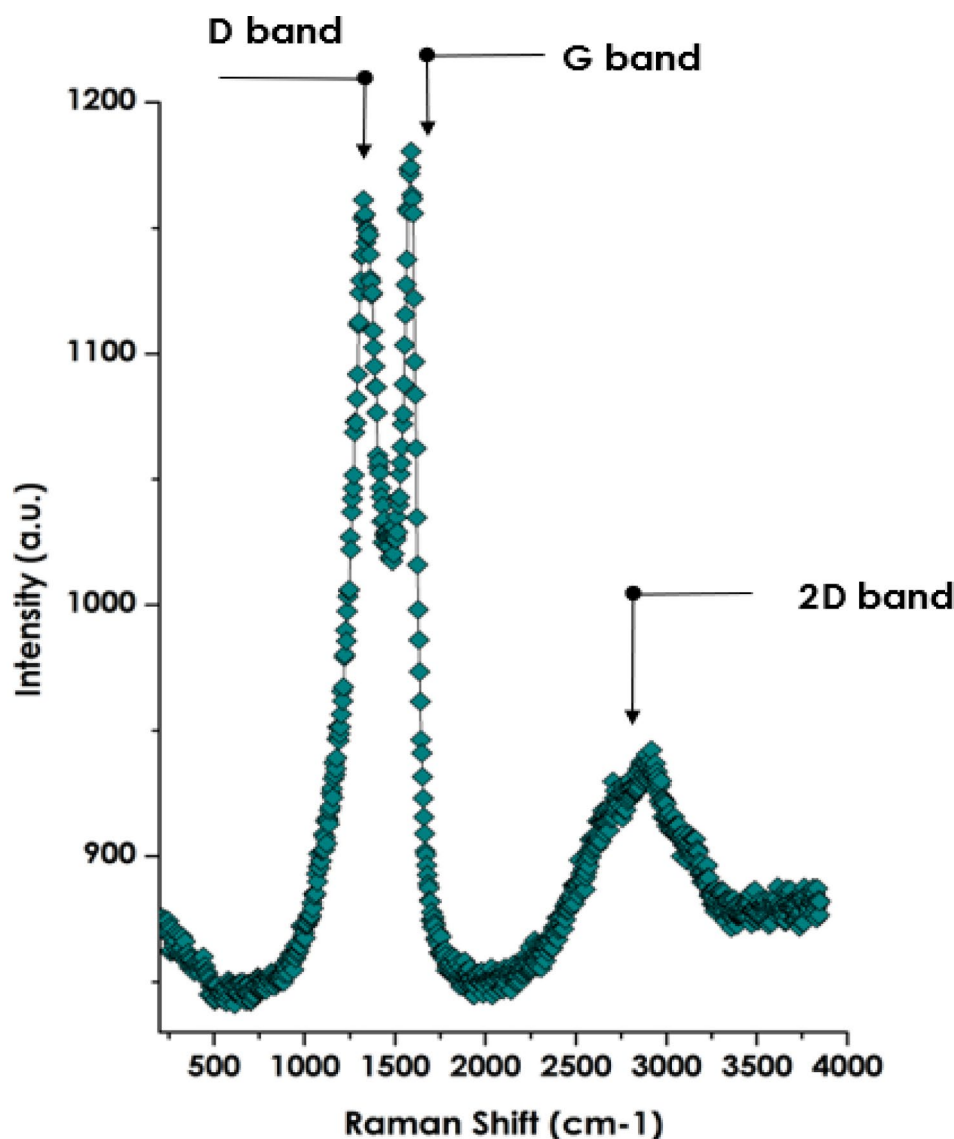


Fig. 7. Raman spectrum of the WS-K₂CO₃ 800 °C porous carbon derived from walnut shells. The spectrum shows the characteristic vibrational bands of carbon materials. The D-band at ~1350 cm⁻¹ indicates structural defects and disorder in the carbon framework. The G-band at ~1580 cm⁻¹ corresponds to graphitic domains and sp²-hybridized carbon.

UV–VIS studies and dye adsorption investigations

WS-K₂CO₃ 800 °C was further evaluated for its adsorption performance against methylene blue dye at varying concentrations. Figure 8 displays the UV–VIS absorbance spectra of MB solutions at initial concentrations of 20 ppm and 40 ppm, monitored over 1 h at room temperature. In Fig. 8a, the 20 ppm MB solution undergoes complete decolourization within 30 min, indicating rapid and effective adsorption by WS-K₂CO₃ 800 °C of MB. This swift uptake reflects the material's high surface area and hierarchical pore structure, which facilitates efficient dye diffusion and binding⁴⁹. Figure 8b demonstrates that WS-K₂CO₃ 800 °C also exhibits strong adsorption capacity at the higher concentration of 40 ppm, although the rate of removal is comparatively slower. This behaviour is consistent with adsorption kinetics, where increased solute concentration imposes a greater demand on available active sites, thereby extending the equilibrium time⁴¹. Nonetheless, the material maintains substantial removal efficiency, underscoring its robustness across a range of dye loadings. These results confirm that WS-K₂CO₃ 800 °C bio-engineered carbon possesses both high affinity and capacity for Methylene Blue adsorption, making it a promising candidate for dye-contaminated wastewater treatment.

WS-K₂CO₃ 800 °C, was also evaluated for its adsorption performance against Methyl Orange (MO) dye at varying concentrations. Figure 9 presents the UV–VIS absorbance spectra of MO solutions at initial concentrations of 20 ppm and 40 ppm, monitored over 1 h at room temperature. In contrast, the adsorption of methyl orange, a negatively charged azo dye, revealed a more gradual uptake profile. Figure 9 illustrates the UV–VIS absorption spectra of MO solutions at initial concentrations of 20 ppm (Fig. 9a) and 40 ppm (Fig. 9b) over

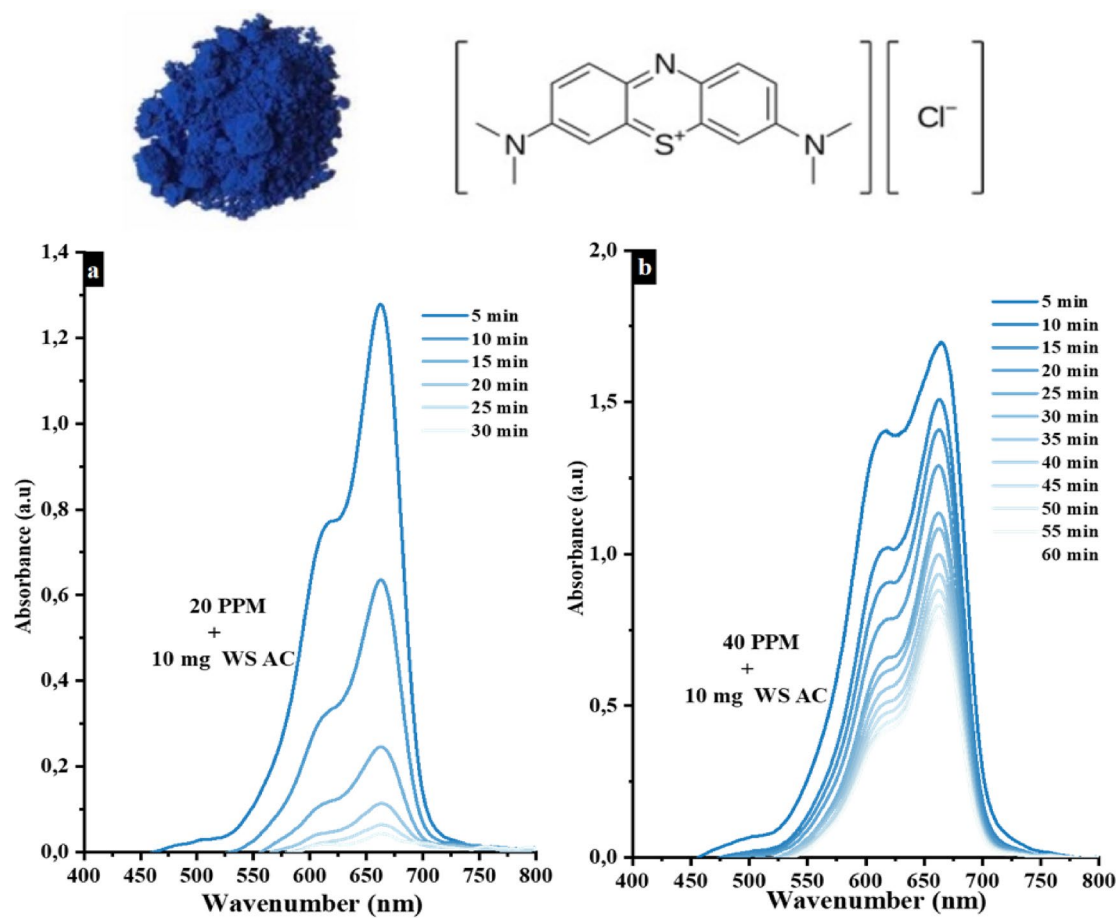


Fig. 8. Adsorption of Methylene Blue onto WS- K_2CO_3 800 °C activated carbon. The plots show the removal of MB over time at different initial concentrations. (a) Adsorption kinetics at 20 ppm MB, showing a rapid decline of the highest peak over time. (b) Adsorption kinetics at 40 ppm MB, showing a slower decrease with higher equilibrium capacity.

1 h. At 20 ppm, MO decolourization occurred rapidly, indicating efficient interaction with WS- K_2CO_3 800 °C. This swift removal is attributed to the material's high surface area and well-developed pore structure, which promote effective dye diffusion and physical entrapment³². At 40 ppm, the rate of MO removal was comparatively slower, reflecting the increased competition for active sites and extended equilibrium time typical of higher solute concentrations. Nonetheless, the material maintained substantial removal efficiency, demonstrating its robustness across a range of dye loadings. The adsorption of MO is likely governed by a combination of hydrogen bonding, van der Waals forces, and pore-filling mechanisms, rather than strong electrostatic interactions^{60,70}.

To further assess the versatility of WS- K_2CO_3 800 °C, adsorption tests were conducted using Congo red (CR), an anionic diazo dye known for its structural complexity and resistance to biodegradation. Figure 10 illustrates the UV-VIS absorption spectra of CR solutions at initial concentrations of 10 ppm (Fig. 10a) and 20 ppm (Fig. 10b), monitored over time at room temperature. At 10 ppm, WS- K_2CO_3 800 °C facilitated rapid decolourization, with a marked reduction in absorbance observed within the first 30 min. This suggests efficient dye diffusion and strong affinity between CR molecules and the adsorbent surface, likely mediated by hydrogen bonding and van der Waals interactions. The material's hierarchical pore structure and surface polarity appear to play a critical role in accommodating the bulky CR molecules and enhancing contact with active sites. At 20 ppm, the adsorption rate was comparatively slower, reflecting increased competition for binding sites and potential steric hindrance due to CR's larger molecular footprint²⁹. Nevertheless, the material maintained substantial removal efficiency, indicating its capacity to handle elevated dye loads without significant performance loss⁷¹. The consistent uptake across both concentrations reinforces WS- K_2CO_3 800 °C's adaptability to structurally diverse dyes. Unlike methyl orange, which relies more on pore accessibility, and methylene blue, which benefits from electrostatic attraction, Congo red adsorption appears to be governed by a combination of physical entrapment and surface interactions, underscoring the multifaceted nature of the adsorbent's functionality.

Effect of solution pH and adsorbent dosage on dye adsorption

The effects of solution pH and adsorbent dosage on dye adsorption were investigated, with results shown in Fig. 11. The initial dye concentrations were fixed at 20 ppm for methylene blue (MB) and methyl orange (MO), and 10 ppm for Congo red (CR). As shown in Fig. 11a and b, the adsorption of MB onto WS- K_2CO_3 -800 °C

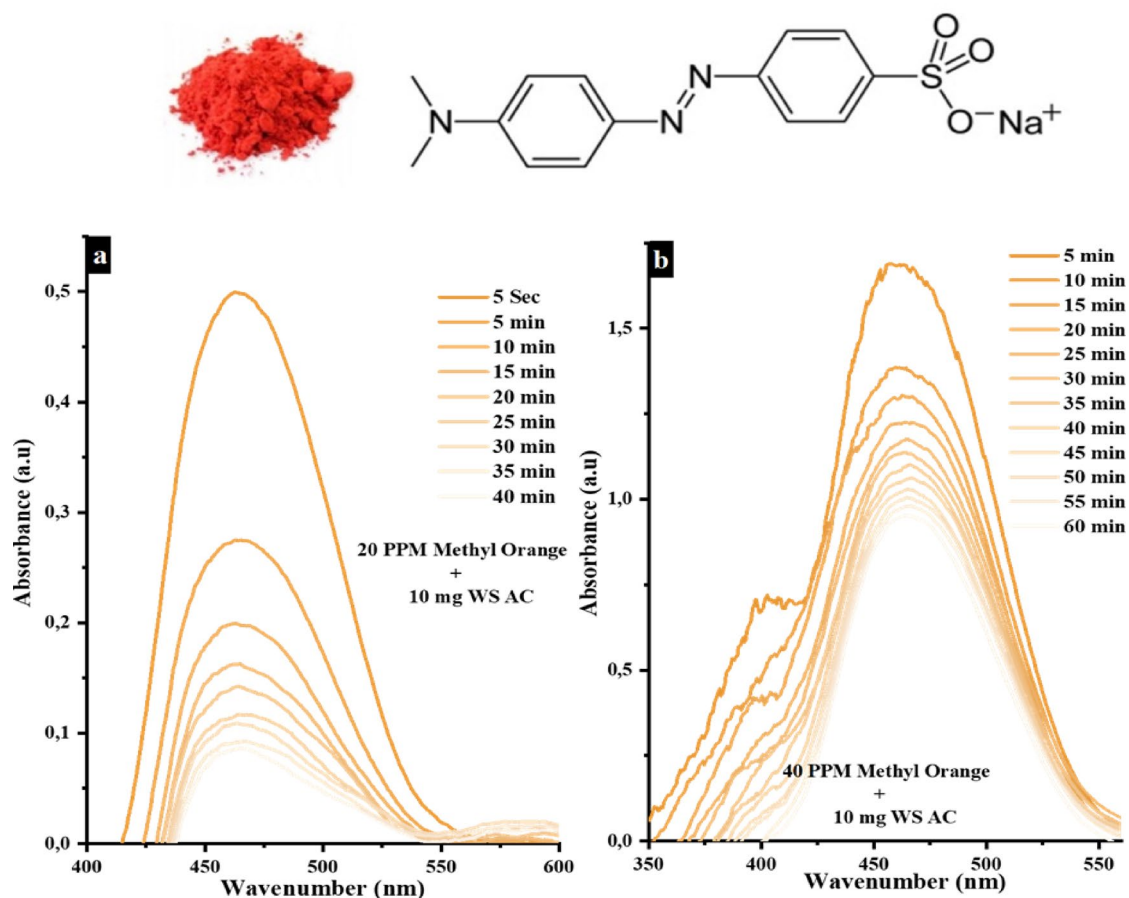


Fig. 9. Adsorption of Methyl Orange onto WS- K_2CO_3 800 °C activated carbon at different concentrations. The plots show the change in MO concentration over time. (a) 20 ppm MO, exhibiting a rapid decline of the highest peak and fast initial uptake. (b) 40 ppm MO, showing a slower decrease but reaching a higher equilibrium adsorption capacity.

activated carbon exhibits minimal pH dependence over the investigated range (pH 4.5–9). This suggests that MB uptake is primarily governed by pore-filling and π - π interactions between the aromatic dye molecules and the carbon surface, while electrostatic effects are not dominant within this pH range. For methyl orange (Fig. 11c,d), adsorption is slightly enhanced under neutral to alkaline conditions (pH 7–9), whereas incomplete removal is observed at pH 4.5. This behavior indicates that dye speciation and proton competition effects influence adsorption under acidic conditions, consistent with reported trends for anionic dyes on carbonaceous adsorbents³². The adsorption of Congo red (Fig. 11e) shows improved removal at pH 4.5 and 9, with reduced efficiency at pH 7, indicating the involvement of multiple adsorption mechanisms. Given the aromatic structure of CR, π - π interactions and hydrogen bonding are likely contributors, while electrostatic interactions alone cannot account for the observed pH-dependent behavior. The effect of adsorbent dosage is presented in Fig. 11g. Increasing the mass of WS- K_2CO_3 -800 °C activated carbon leads to a progressive increase in dye removal due to the greater availability of adsorption sites. Complete removal of all dyes was achieved at an adsorbent dosage of 20 mg, demonstrating the high adsorption efficiency of the prepared material.

Kinetics study

The adsorption behavior of methylene blue (MB), methyl orange (MO), and Congo red (CR) onto WS- K_2CO_3 -800 °C AC was evaluated through kinetic modeling and removal efficiency analysis, with results illustrated in Fig. 12 and Table 3. The kinetic experiments were conducted at initial dye concentrations of 40 ppm for MB and MO and 20 ppm for CR, enabling direct comparison of adsorption dynamics across dyes with distinct molecular structures.

As shown in Fig. 12a, MB adsorption proceeds rapidly during the initial contact period, followed by a gradual approach to equilibrium. Both pseudo-first-order (PFO) and pseudo-second-order (PSO) models provide excellent fits to the experimental data, with high correlation coefficients ($R^2 \geq 0.995$). However, the PSO model yields the highest R^2 value (0.998) and an equilibrium adsorption capacity ($Q_e = 185.8 \text{ mg g}^{-1}$) that closely matches the experimental uptake. This indicates that MB adsorption is primarily governed by surface-related processes^{25,31}. The relatively low correlation obtained from the intraparticle diffusion (IPD) model ($R^2 = 0.567$),

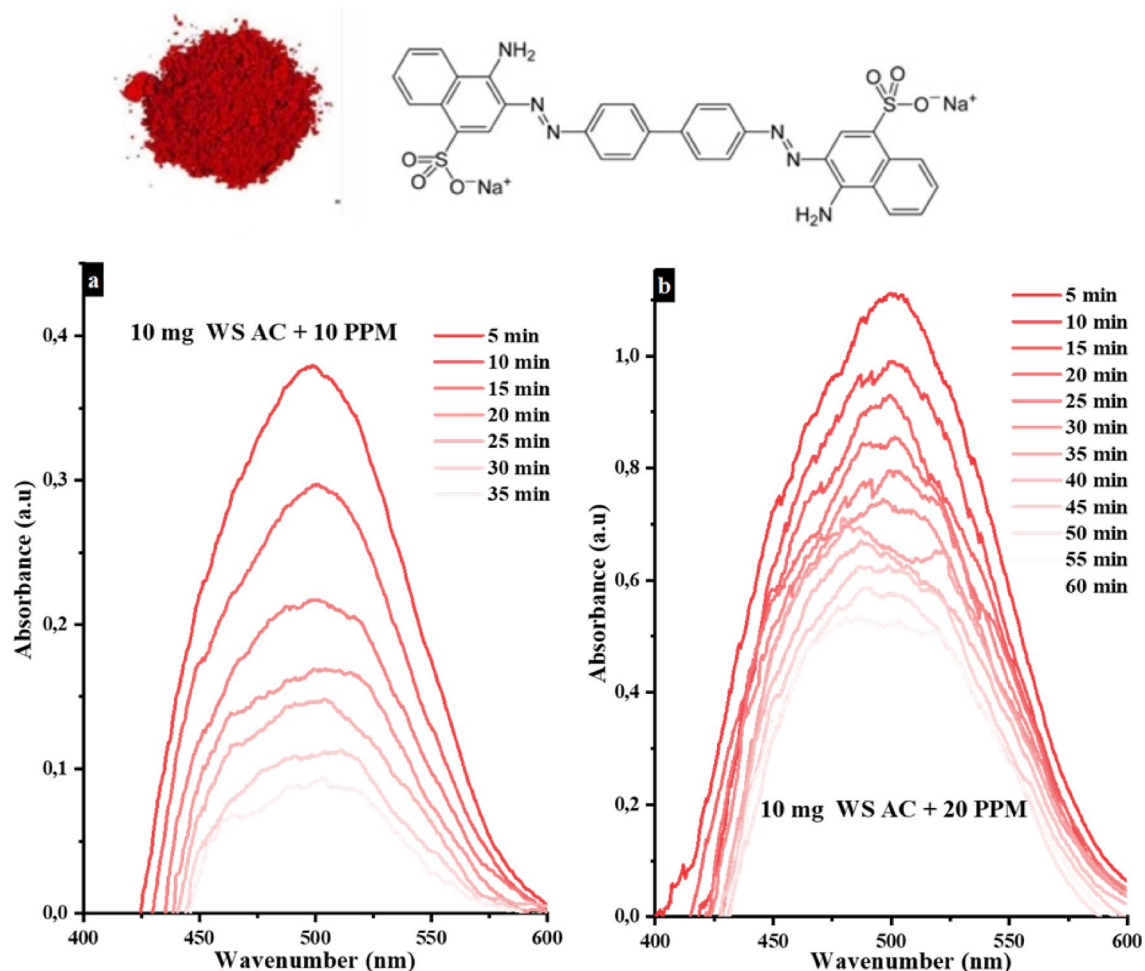


Fig. 10. Adsorption of Red Congo onto WS- K_2CO_3 800 °C activated carbon at different concentrations. The plot shows the change in dye concentration over time. **(a)** 10 ppm Red Congo, showing rapid initial uptake and a steep decline of the highest peak. **(b)** 20 ppm Red Congo, exhibiting a slight decrease but higher equilibrium adsorption.

together with a large intercept value ($C = 79.37 \text{ mg g}^{-1}$), suggests that intraparticle diffusion contributes to the overall process but is not the sole rate-limiting step.

A comparable kinetic trend is observed for methyl orange (Fig. 12b). The PSO model again provides the best fit ($R^2 = 0.996$), outperforming the PFO model ($R^2 = 0.975$), with a calculated Q_e value of 133.72 mg g^{-1} . These results indicate that MO adsorption is also dominated by surface-controlled interactions^{32,72}. The moderate fit of the IPD model ($R^2 = 0.838$) and its non-zero intercept ($C = 32.62 \text{ mg g}^{-1}$) imply that diffusion-related processes coexist with surface adsorption, particularly at later stages of uptake.

In contrast, the adsorption kinetics of Congo red (Fig. 12c) are best described by the intraparticle diffusion model, which exhibits the highest correlation coefficient ($R^2 = 0.996$). This highlights the dominant role of diffusion-controlled transport in CR adsorption. The comparatively lower diffusion rate constant ($k_{id} = 8.01 \text{ mg g}^{-1} \text{ min}^{-1/2}$) and the near-zero intercept ($C = -1.89 \text{ mg g}^{-1}$) suggest minimal boundary-layer resistance and emphasize the influence of internal pore diffusion^{45,71}. Although both PFO and PSO models also show high R^2 values (> 0.99), the superior IPD fit reflects the larger molecular size and structural complexity of CR, which can hinder rapid surface adsorption and favor diffusion-limited uptake.

The comparative removal efficiencies presented in Fig. 12d further corroborate these findings, with MB exhibiting the fastest and most efficient removal, followed by MO and CR. These differences can be attributed to variations in molecular size, diffusion behavior, and accessibility of adsorption sites.

Overall, the kinetic analysis demonstrates that adsorption on WS- K_2CO_3 -800 °C activated carbon follows a mixed kinetic mechanism, with PSO-dominated behavior for MB and MO and diffusion-controlled adsorption for CR. This versatility highlights the effectiveness of the prepared activated carbon for the removal of dyes with diverse physicochemical characteristics and adsorption pathways.

Across all dyes, adsorption follows pseudo-second-order kinetics, indicating that chemisorption and active site availability govern uptake. The consistent and strong performance across cationic, anionic, and diazo dyes demonstrates that WS K_2CO_3 800 °C is a versatile, broad-spectrum adsorbent. Its hierarchical pore network and

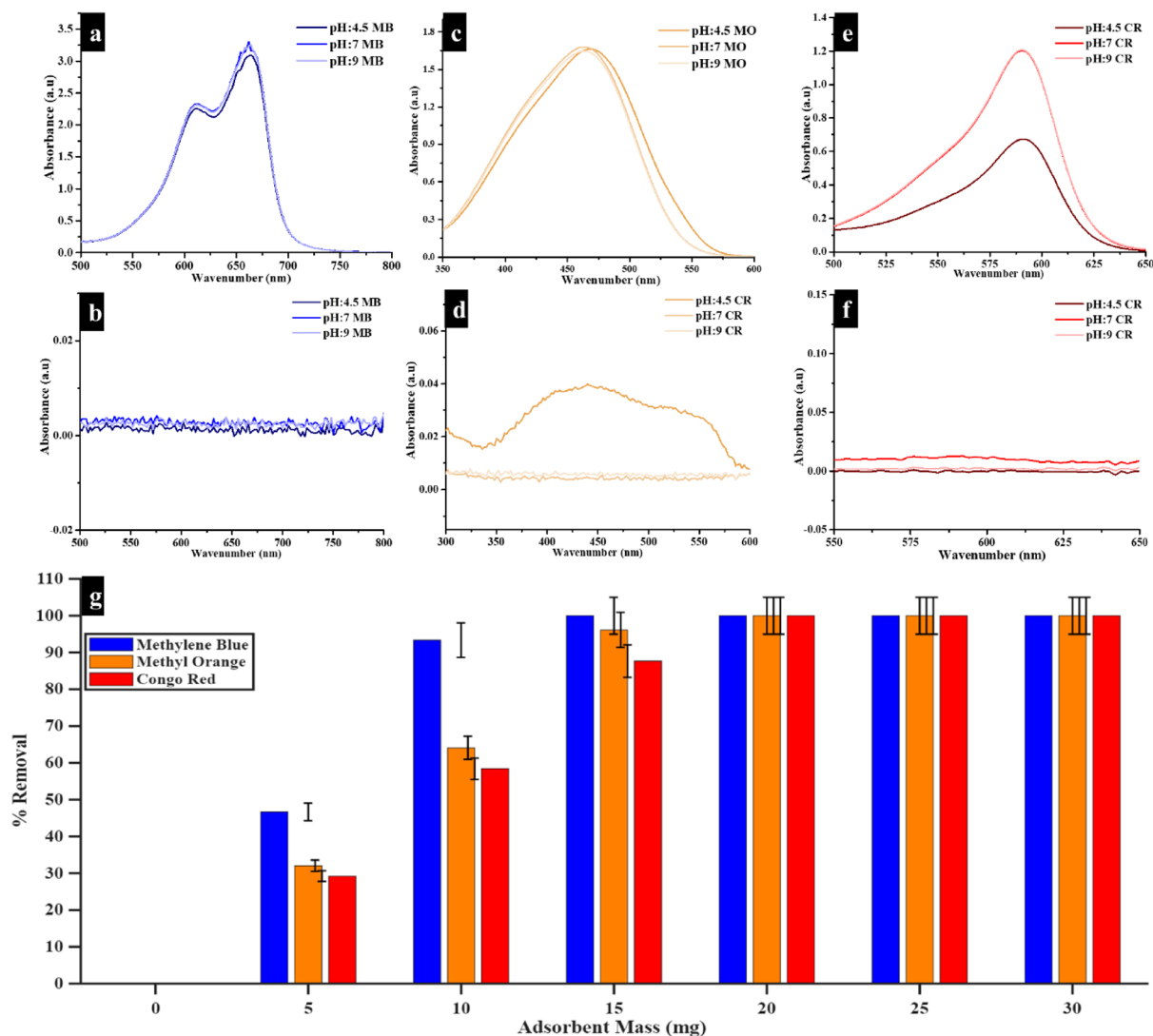


Fig. 11. Effect of solution pH and adsorbent dosage on dye adsorption by WS-K₂CO₃-800 °C activated carbon. (a,b) UV-VIS spectra of methylene blue before and after adsorption at pH 4.5, 7, and 9 (C₀ = 20 ppm); (c,d) UV-VIS spectra of methyl orange before and after adsorption at pH 4.5, 7, and 9 (C₀ = 20 ppm); (e,f) UV-VIS spectra of Congo red before and after adsorption at pH 4.5, 7, and 9 (C₀ = 10 ppm); (g) Effect of adsorbent dosage on dye removal efficiency.

high surface area promote efficient diffusion and strong dye-surface interactions, enabling capacities comparable to or superior to many reported biomass-derived activated carbons.

The adsorption performance obtained in this study is compared with previously reported results in Table 4.

The adsorption equilibrium of methylene blue (MB), methyl orange (MO), and Congo red (CR) onto WS-K₂CO₃-800 °C activated carbon was evaluated using the Langmuir, Freundlich, and Temkin isotherm models (Fig. 13), and the corresponding parameters are gathered in Table 5.

For methylene blue (Fig. 13a), none of the tested models provided an excellent description of the experimental data. The Langmuir model exhibited a relatively low correlation coefficient (R^2 : 0.728), suggesting that monolayer adsorption on a homogeneous surface is not fully applicable²⁷. The Freundlich model showed a negligible correlation (R^2 : 0), indicating poor representation of surface heterogeneity effects. The Temkin model yielded a slightly improved fit (R^2 : 0.772), implying that adsorbate-adsorbent interactions and a gradual decrease in adsorption heat may contribute to the adsorption mechanism, consistent with mixed physical-chemical interactions⁴¹.

In contrast, methyl orange (Fig. 13b) adsorption was best described by the Freundlich isotherm (R^2 : 0.825), indicating multilayer adsorption on a heterogeneous surface. The Freundlich constant $n=1.54$ confirms favourable adsorption, while the lower performance of the Langmuir model (R^2 : 0.776) suggests that MO adsorption does not follow ideal monolayer behavior⁷⁰. The Temkin model showed moderate agreement (R^2 : 0.742), further supporting the role of adsorbate-adsorbent interactions rather than uniform site energies.

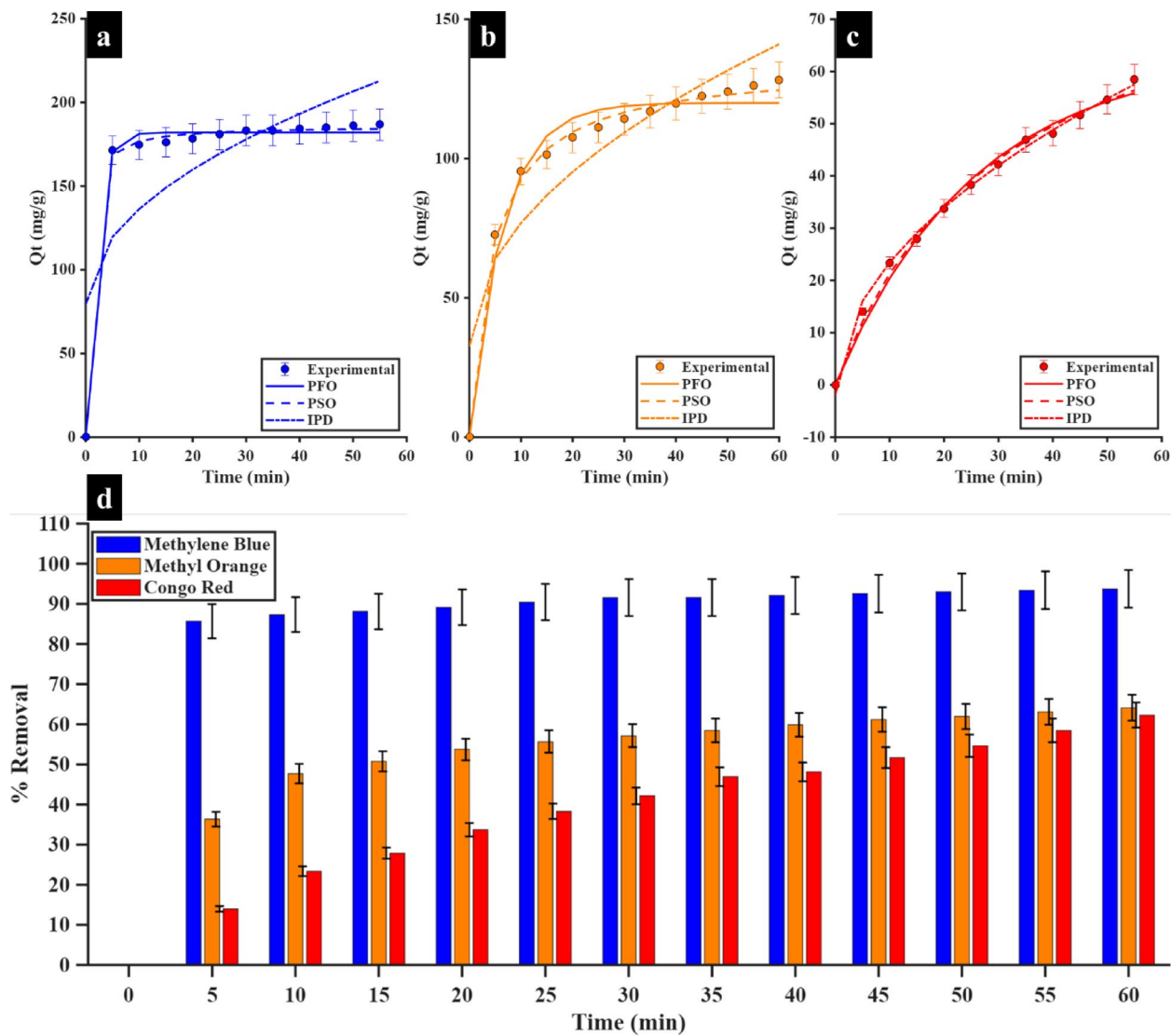


Fig. 12. Adsorption kinetics of methylene blue, methyl orange, and Congo red on WS-K₂CO₃-800 °C activated carbon.

Kinetic model	Parameter	Methylene Blue	Methyl Orange	Congo Red
<i>Pseudo-first order (PFO)</i> $\frac{dQ}{dt} = k_1 (Q_e - Q_t) \therefore Q_t = Q_e (1 - e^{-k_1 t})$	Q_e (mg g ⁻¹)	181.9	120	63.28
	k_1 (min ⁻¹)	0.56	0.15	0.04
	R^2	0.995	0.975	0.991
<i>Pseudo-second order (PSO)</i> $\frac{dQ_t}{dt} = k_2 (Q_e - Q_t)^2 \therefore Q_t = \frac{k_2 Q_e^2 t}{1 + k_2 Q_e t}$	Q_e (mg g ⁻¹)	185.8	133.72	88.94
	k_2 (g mg ⁻¹ min ⁻¹)	0.011	0.00170	3.54×10^{-4}
	R^2	0.998	0.996	0.995
<i>Intraparticle diffusion (IPD)</i> $Q_t = k_{id} t^{\frac{1}{2}} + C$	k_{id} (mg g ⁻¹ min ^{-1/2})	17.99	14.00	8.01
	C (mg g ⁻¹)	79.37	32.62	-1.89
	R^2	0.567	0.838	0.996

Table 3. Kinetic model parameters for the adsorption of methylene blue, methyl orange, and Congo red on WS-K₂CO₃-800 °C activated carbon.

For Congo red (Fig. 13c), the Freundlich model again provided the best fit ($R^2=0.866$), highlighting strong surface heterogeneity and multilayer adsorption behavior. The relatively high n value (3.52) reflects favorable adsorption at low equilibrium concentrations. The Langmuir model displayed poor correlation ($R^2: 0.503$), indicating that CR adsorption does not conform to monolayer assumptions, likely due to its large molecular size

Adsorbent/activation method	Dye	Experimental conditions	q _e /q _m (mg g ⁻¹)	Best-fit kinetic model	Mechanistic notes	Refs.
WS K ₂ CO ₃ 800 °C	MB	20–40 ppm, 25 °C	96.2; 181.8	PSO	Strong affinity via electrostatic attraction + π–π interactions; excellent diffusion due to hierarchical pores	This work
WS K ₂ CO ₃ 800 °C	MO	20–40 ppm	88.8; 115	PSO	Π–π stacking with graphitic domains; H-bonding supports uptake despite anionic charge	This work
WS K ₂ CO ₃ 800 °C	CR	10–20 ppm	36.7; 90.7	PSO	Large azo dye; intraparticle diffusion dominates at higher loading	This work
Pomegranate fruit peel adsorbent	MB	25–300 mg L ⁻¹	218–197.45	PFO	Moderate porosity; fast uptake; eco-friendly precursor	30
AC–ZnO–NH ₃ composite	MB	10–100 mg L ⁻¹	≈ 2.45	PSO	Very low adsorption; limited active sites	27
Chitosan/activated carbon composite	MB	10–100 mg L ⁻¹	15–19	PSO	High stability, low capacity	25
Kitchen-waste biochar	MO	5–25 mg L ⁻¹	22.5	Langmuir	Low SSA; low MO affinity	32
Boiler residue AC	MO	10–150 mg L ⁻¹	141.42; 107.9; 55.86	PSO	Strong MO binding across wide concentrations range	73
Chitosan-modified AC	MO	20–200 mg L ⁻¹	168	PSO	Protonated amines greatly enhance MO uptake	74
Chitosan-added watermelon pulp	CR	10–100 mg L ⁻¹	490.1	PSO	Extremely high CR uptake due to abundant amine groups	36
Sorghum husk biosorbent	CR	10–210 mg L ⁻¹	77.14; 57.55	PSO	Sustainable, moderate CR capacity	35
Guava leaf activated carbon	CR	10–50 mg L ⁻¹	43.35	PFO	Limited pore structure; low CR uptake	71

Table 4. Comparison of the optimized WS-K₂CO₃ 800 °C bio-engineered carbon with previously reported literature.

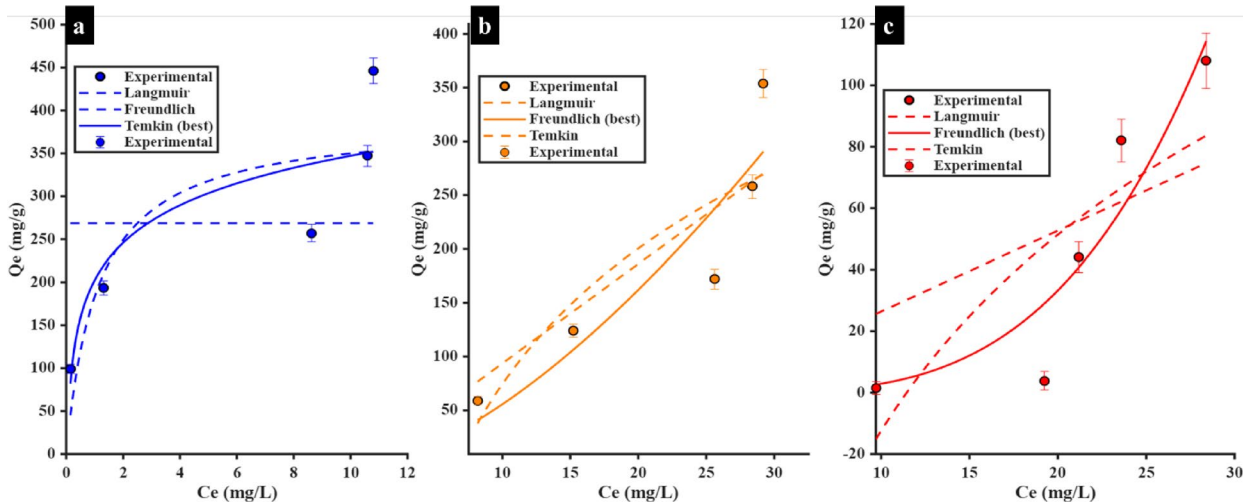


Fig. 13. Adsorption isotherms of (a) methylene blue, (b) methyl orange, and (c) Congo red onto WS-K₂CO₃-800 °C activated carbon fitted with Langmuir, Freundlich, and Temkin models.

Isotherm model	Parameter	Methylene Blue	Methyl Orange	Congo Red
Langmuir $\frac{C_e}{Q_e} = \frac{1}{q_m K_e} + \frac{C_e}{q_m}$	Q _{max} (mg g ⁻¹)	389.21	15,527.76	9628.51
	KL (L mg ⁻¹)	0.8898	6.07 × 10 ⁻⁴	2.75 × 10 ⁻⁴
	R ²	0.728	0.776	0.503
Freundlich $Q_e = K_F C_e^{1/n}$	K _F ((mg g ⁻¹) (L mg ⁻¹) ^{1/n})	268.50	1.59	8.70 × 10 ⁻⁴
	n	~*	1.54	3.52
	R ²	~ 0	0.825	0.866
Temkin $Q_e = \frac{Rt}{b} \ln K_T + \frac{Rt}{b} \ln C_e$	B (J mol ⁻¹)	62.91	181.92	92.02
	A (L g ⁻¹)	24.92	0.150	0.087
	R ²	0.772	0.742	0.640

Table 5. Isotherm model parameters and correlation coefficients for the adsorption of methylene blue, methyl orange, and Congo red onto WS-K₂CO₃-800 °C activated carbon.

Dye	Adsorbent	$Q_{\max}$ (mg g ⁻¹)	Best-fitting isotherm	Reference
Methylene Blue	Walnut shell powder	~ 315	Langmuir	24
	Citrate-modified bagasse	~ 298	Langmuir	37
	Almond & walnut shell AC	~ 340	Langmuir	47
	WS-K ₂ CO ₃ -800 °C AC (this work)	389.2	Temkin-dominant	This work
Methyl Orange	Pomelo peel activated biochar	~	Freundlich	70
	Aloe vera waste adsorbent	~	Freundlich	72
	Kitchen-waste biochar	~	Freundlich	32
	WS-K ₂ CO ₃ -800 °C AC (this work)	~	Freundlich	This work
Congo Red	Guava-leaf activated carbon	~	Freundlich	71
	Chitosan–watermelon pulp biosorbent	~	Freundlich	36
	Amine-functionalized walnut shell	~	Freundlich	40
	WS-K ₂ CO ₃ -800 °C AC (this work)	~	Freundlich	This work

Table 6. Comparison of adsorption isotherm performance of WS-K₂CO₃-800 °C activated carbon with selected literature adsorbents.

and complex aromatic structure²⁹. The Temkin model showed moderate fitting (R^2 : 0.640), suggesting a non-uniform distribution of adsorption energies³⁶.

Overall, the isotherm analysis demonstrates that adsorption of MO and CR on WS-K₂CO₃-800 °C AC is predominantly governed by heterogeneous, multilayer adsorption mechanisms, whereas MB adsorption follows a more complex behavior that cannot be adequately captured by a single classical isotherm model.

To contextualize the adsorption performance of WS-K₂CO₃-800 °C activated carbon, its equilibrium parameters were benchmarked against representative adsorbents reported in the literature for methylene blue, methyl orange, and Congo red removal. Table 6 compares the maximum adsorption capacities, best-fitting isotherm models, and correlation coefficients, demonstrating that the present material exhibits adsorption performances that are comparable to or exceed those of several previously reported biomass- and waste-derived adsorbents, while maintaining consistent isotherm behavior and strong model fitting.

As follow-up studies, it is intended to optimise the conditions of (1) fast removal of the investigated dyes. (2) effect of the solutions' pH, (3) extension to other natural and synthetic dyes and pollutants, (4) combining the current bio-engineered active carbon with other simple or multi-compound photocatalysts^{19,75–81}

Conclusion

Walnut-shell-derived activated carbon prepared via K₂CO₃ activation at 800 °C (WS-K₂CO₃-800 °C) proved to be a sustainable and highly efficient adsorbent for dye-contaminated wastewater. The material exhibited a high surface area (1786.64 m² g⁻¹) and hierarchical porosity, enabling rapid uptake of methylene blue, methyl orange, and Congo red, with near-complete removal achieved within 30 min at low concentrations. Kinetic modeling confirmed that adsorption followed the pseudo-second-order model ($R^2 > 0.99$), indicating chemisorption as the dominant mechanism, while the intraparticle diffusion model suggested multi-stage uptake involving surface binding and pore diffusion. Equilibrium capacities reached 185.8 mg g⁻¹ (MB), 133.7 mg g⁻¹ (MO), and 88.9 mg g⁻¹ (CR). Isotherm analyses revealed heterogeneous and multilayer adsorption, with the Freundlich and Temkin models providing the best fits. These findings highlight walnut shells as a low-cost, renewable precursor for advanced porous carbons, offering scalable and environmentally friendly solutions for wastewater remediation.

Data availability

All datasets generated or analyzed in this study are provided within the published article and its Supplementary Information. Additional data can be obtained from the corresponding author upon reasonable request.

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Declarations

Competing interests

The authors declare no competing interests.

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